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Chemical Bonding and Hybridisation — Issue #32 specimen: Source questions

Q1 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Hybridisation, orbital conservation, and overlap symmetry

Concept difficulty: Medium concept

Question difficulty: Direct application

Family: Hybrid-orbital mapping, orbital symmetry, and 3D stereochemical representation

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

In the introductory valence-bond model,
name the hybrid-orbital set used for four

Representation

equivalent electron-domain directions around a central atom, and state how many hybrid orbitals it contains.

Conditions

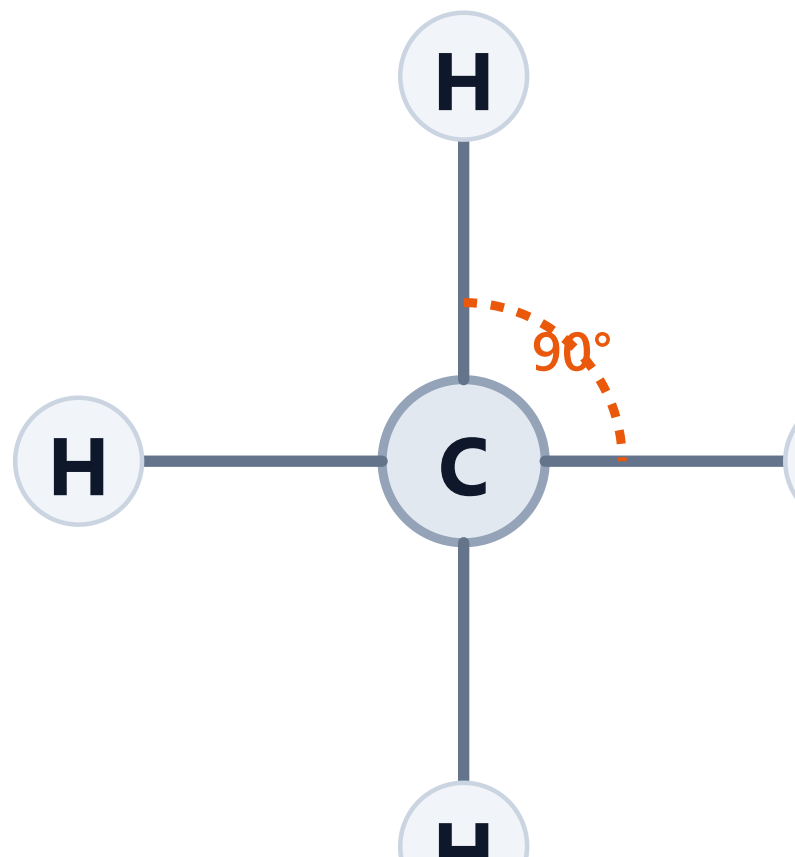
- Introductory valence-bond model and VSEPR rules apply; assume ideal coordination geometry unless specified.

Common wrong route · open if you want a warning

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Stage 1: 2D Lewis Cross Sket

Topological connectivity (Flat 90° illusio





Misleading: 2D drawing forces crowded 9

Flat 2D cross

Need a hint? · guided support



Need the concept again?

Hybridisation, orbital conservation, and overlap symmetry

Q2 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Steric electron-domain counting and multiple bonds

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Steric electron-domain counting and multiple-bond constraints

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

The Lewis structure $\text{H}_2\text{C}=\text{O}$ has a double bond from carbon to oxygen and one single bond from carbon to each hydrogen. Explain why the $\text{C}=\text{O}$ double bond counts as one electron domain around carbon.

Conditions

- Introductory valence-bond model and VSEPR rules apply; assume ideal coordination geometry unless specified.

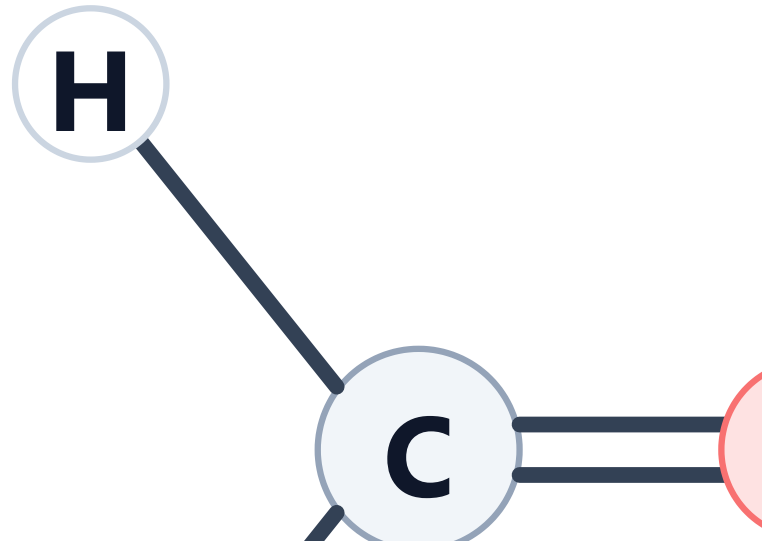
Common wrong route · open if you want a warning

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Representation

Stage 1: Lewis Form

Four shared electron pairs





Lewis structure

Need a hint? • guided support



Need the concept again?

Steric electron-domain counting and multiple bonds

Q3 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Hybridisation, orbital conservation, and overlap symmetry

Concept difficulty: Medium concept

Question difficulty: Direct application

Family: Hybrid-orbital mapping, orbital symmetry, and 3D stereochemical representation

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

For CH₄, use the known rule for four bonding domains and no central lone pair to assign the carbon hybridisation and electron-domain geometry. Show the domain count.

Representation

Conditions

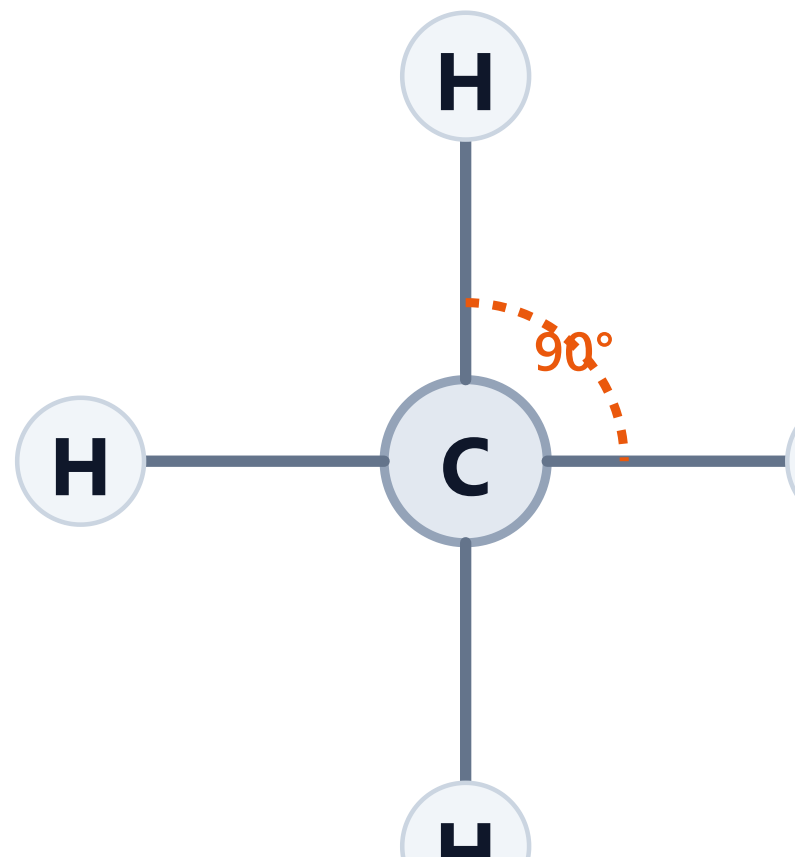
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Common wrong route • open if you want a warning

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Stage 1: 2D Lewis Cross Sket

Topological connectivity (Flat 90° illusio





Misleading: 2D drawing forces crowded 9

Flat 2D cross

Need a hint? · guided support



Need the concept again?

Hybridisation, orbital conservation, and overlap symmetry

Q4 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: 3D Stereochemical representation and qualitative models

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Hybrid-orbital mapping, orbital symmetry, and 3D stereochemical representation

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

NH₃ has three N–H single bonds and one lone pair on nitrogen. Translate that information into a labelled diagram showing all four electron domains; distinguish the lone pair from the three bond directions. Do not treat the planar Lewis sketch as the molecular shape.

Conditions

- Introductory valence-bond model and VSEPR rules apply; assume ideal coordination geometry unless specified.

Representation

NH₃: three N–H bonds; one N l

N

Inventory

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Need the concept again?

3D Stereochemical representation and qualitative models

Q5 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: 3D Stereochemical representation and qualitative models

Concept difficulty: Medium concept

Question difficulty: Multi-step reasoning

Family: Hybrid-orbital mapping, orbital symmetry, and 3D stereochemical representation

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

A model is needed only to describe the overall three-dimensional arrangement of electron domains in NH_3 ; it need not

Representation

describe the detailed orbital overlap.
Choose between VSEPR and a hybrid-orbital overlap diagram, and identify what your chosen model predicts and what it leaves unresolved.

Conditions

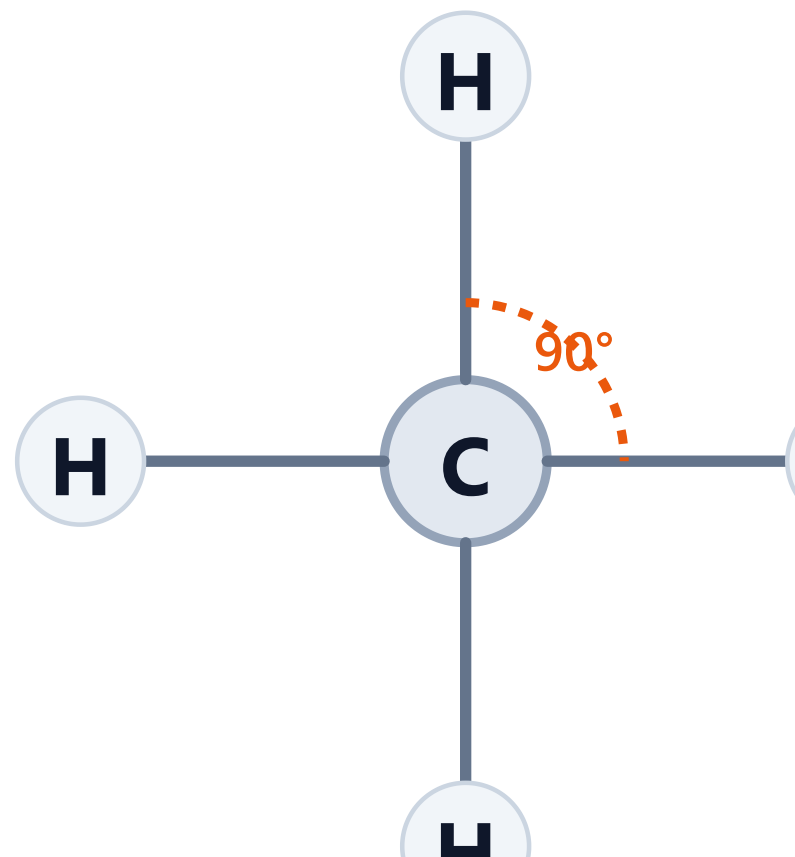
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Stage 1: 2D Lewis Cross Sket

Topological connectivity (Flat 90° illusio





Misleading: 2D drawing forces crowded 9

Flat 2D cross

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Need the concept again?

3D Stereochemical representation and qualitative models

Q6 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Hybridisation, orbital conservation, and overlap symmetry

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Hybrid-orbital mapping, orbital symmetry, and 3D stereochemical representation

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

For a qualitative orbital sketch, overlap A is end-on along the internuclear axis and overlap B is side-on on opposite sides of that axis. Classify each as sigma or pi overlap and identify the defining feature you used.

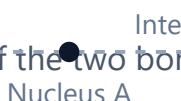
Conditions

- Introductory valence-bond model and VSEPR rules apply; assume ideal coordination geometry unless specified.

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Representation

Stage 1: Internuclear Axis Reference

The internuclear axis (z-axis) connects the nuclei of the two boron atoms.  Inte
Nucleus A

Internuclear axis

Need a hint? • guided support

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Need the concept again?

Hybridisation, orbital conservation, and overlap symmetry

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Q7 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: 3D Stereochemical representation and qualitative models

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Hybrid-orbital mapping, orbital symmetry, and 3D stereochemical representation

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

For CH₄, sketch a three-dimensional arrangement of the four C–H bond directions using a wedge/dash convention or another clearly defined convention. Explain how your drawing differs from a flat cross-shaped Lewis sketch.

Representation

Conditions

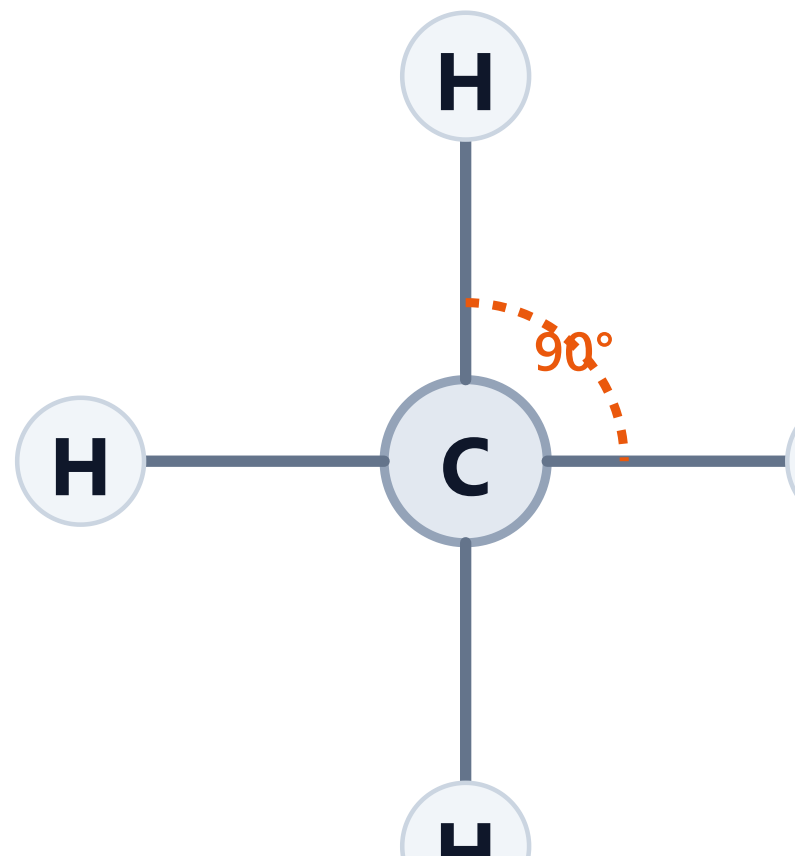
- Introductory valence-bond model and VSEPR rules apply; assume ideal coordination geometry unless specified.

Common wrong route • open if you want a warning

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Stage 1: 2D Lewis Cross Sket

Topological connectivity (Flat 90° illusio





Misleading: 2D drawing forces crowded 9

Flat 2D cross

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Need the concept again?

3D Stereochemical representation and qualitative models

Q8 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Steric electron-domain counting and multiple bonds

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Steric electron-domain counting and multiple-bond constraints

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

A nitrogen atom in NF_3 has three N–F single bonds and one lone pair. Each fluorine also has lone pairs. Count the electron domains around nitrogen and state which lone pairs belong in that central-atom count.

Conditions

- Introductory valence-bond model and VSEPR rules apply; assume ideal coordination geometry unless specified.

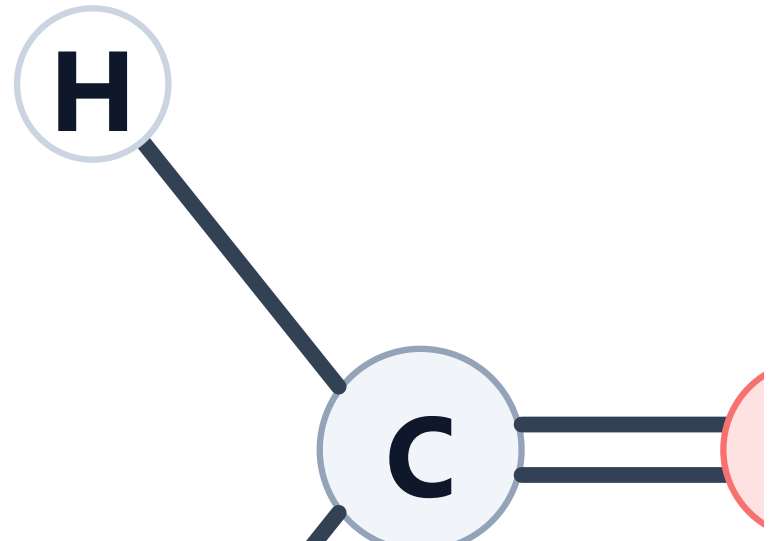
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Representation

Stage 1: Lewis Form

Four shared electron pairs





Lewis structure

Need a hint? • guided support



Need the concept again?

Steric electron-domain counting and multiple bonds

Q9 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Steric electron-domain counting and multiple bonds

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Steric electron-domain counting and multiple-bond constraints

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

A student counts the double bond in $\text{H}_2\text{C}=\text{O}$ as two separate directions and predicts four domains around carbon. Locate the counting error and give the corrected domain count.

Representation

Conditions

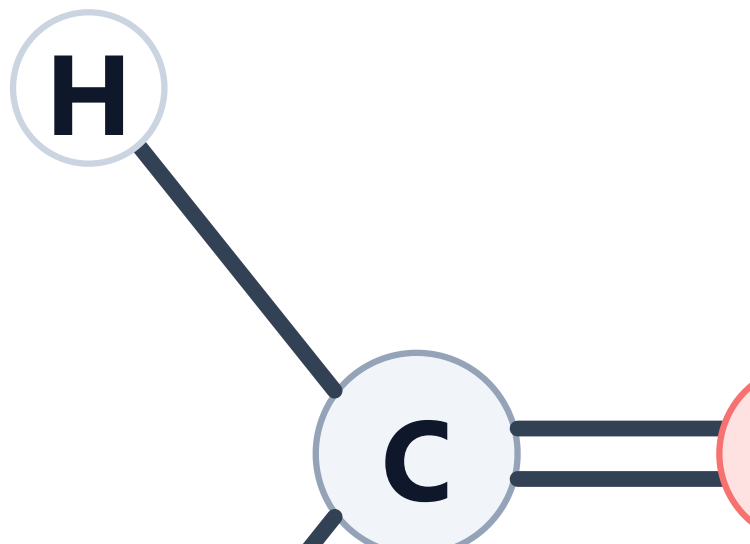
- Introductory valence-bond model and VSEPR rules apply; assume ideal coordination geometry unless specified.

Common wrong route • open if you want a warning

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Stage 1: Lewis Form

Four shared electron pairs





Student trap: counts 4 bonds
as 4 separate directions

Lewis structure

Need a hint? · guided support

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Need the concept again?

Steric electron-domain counting and multiple bonds

Q10 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Hybridisation, orbital conservation, and overlap symmetry

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Hybrid-orbital mapping, orbital symmetry, and 3D stereochemical representation

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

The benchmark supplies this table: BF₃ has three bonding domains and no central lone pair; CH₄ has four bonding domains and

Representation

no central lone pair. Match each central atom to the introductory sp^2 or sp^3 model, and check that the number of hybrid-orbital directions agrees with the domain count.

Conditions

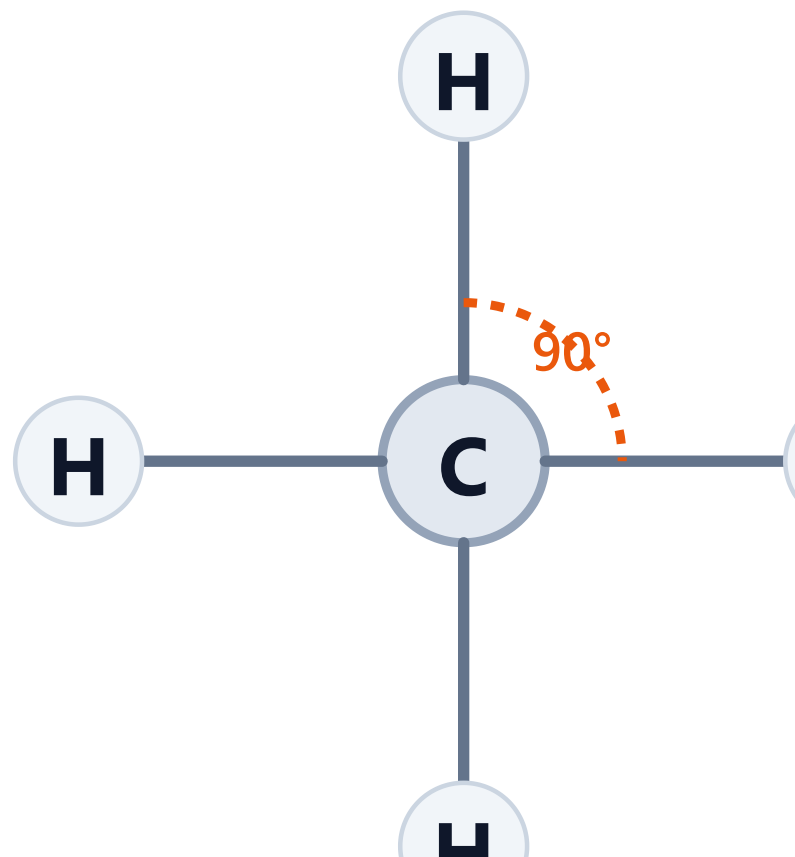
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Stage 1: 2D Lewis Cross Sket

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Need the concept again?

Hybridisation, orbital conservation, and overlap symmetry